KAIZEN MATHS

Lesson starter

Integration by Substitution and Parts

Integration by Substitution

Name _____

S1

Integrate each expression using a linear substitution.

$$\int 8(4x-5)^4\,dx$$

S2

Integrate each expression using a quadratic substitution.

$$\int 3(2x-3)(x^2-3x+3)^2\,dx$$

S3

Integrate each exponential expression using substitution.

$$\int 12e^{4x+2}\,dx$$

S4

Integrate each expression using a linear substitution.

$$\int 20(5x+3)^4\,dx$$

KAIZEN MATHS

Lesson starter

Integration by Substitution and Parts

Integration by Substitution

Name _____

S1

Integrate each expression using a linear substitution.

$\int 8(4x-5)^4\,dx$

S2

Integrate each expression using a quadratic substitution.

$\int 3(2x-3)(x^2-3x+3)^2\,dx$

S3

Integrate each exponential expression using substitution.

$\int 12e^{4x+2}\,dx$

S4

Integrate each expression using a linear substitution.

$\int 20(5x+3)^4\,dx$

ALG-99DB-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Integration by Substitution and Parts

Integration by Substitution

Name _____

EA1

Integrate each expression using a quadratic substitution.

$$\int 3(4x-5)(2x^2-5x-1)^3\,dx$$

EA2

Integrate each exponential expression using substitution.

$$\int 9e^{3x-5}\,dx$$

Show your working

KAIZEN MATHS

Exit ticket

Integration by Substitution and Parts

Integration by Substitution

Name _____

EB1

Integrate each expression using a linear substitution.

$$\int 25(5x+7)^5\,dx$$

Show your working

KAIZEN MATHS

Exit ticket

Integration by Substitution and Parts

Integration by Substitution

Name _____

EA1

Integrate each expression using a quadratic substitution.

$$\int 3(4x-5)(2x^2-5x-1)^3\,dx$$

EA2

Integrate each exponential expression using substitution.

$$\int 9e^{3x-5}\,dx$$

ALG-99DB-L1 · EA

Show your working

KAIZEN MATHS

Exit ticket

Integration by Substitution and Parts

Integration by Substitution

Name _____

EB1

Integrate each expression using a linear substitution.

$$\int 25(5x+7)^5\,dx$$

ALG-99DB-L1 · EB

Show your working

Teacher answers & working

Integration by Substitution and Parts · Integration by Substitution

ALG-99DB-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Integrate each expression using a linear substitution.

$\int 8(4x-5)^4\,dx$

Answer:

$\frac{2}{5}(4x-5)^5+C$

Working:

Let $u=4x-5$.

Differentiate: $\frac{du}{dx}=4$, so $du=4\,dx$ and $dx=\frac{1}{4}\,du$.

Substitute into the integral: $\int 8(4x-5)^4\,dx=\int 8u^4\cdot\frac{1}{4}\,du$.

Simplify the constant: $\int 2u^4\,du$.

New power: $4+1=5$.

Integrate: $2\cdot\frac{u^5}{5}+C$.

Simplify the coefficient: $2/5=\frac{2}{5}$.

$\frac{2}{5}u^5+C$.

Replace u with $4x-5$: $\frac{2}{5}(4x-5)^5+C$.

Teacher answers & working

Integration by Substitution and Parts · Integration by Substitution

ALG-99DB-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S2

Integrate each expression using a quadratic substitution.

$$\int 3(2x-3)(x^2-3x+3)^2\,dx$$

Answer:

$(x^2-3x+3)^3+C$

Working:

Let $u=x^2-3x+3$.

Differentiate: $\dfrac{du}{dx}=2x-3$, so $du=(2x-3)\,dx$.

The factor $(2x-3)\,dx$ appears in the integral, so replace it with du.

The integral becomes $\int 3u^2\,du$.

New power: $2+1=3$.

Integrate: $3\cdot\dfrac{u^3}{3}+C$.

Simplify the coefficient: $3/3=1$.

u^3+C.

Replace u with the original expression: $(x^2-3x+3)^3+C$.

S3

Integrate each exponential expression using substitution.

$$\int 12e^{4x+2}\,dx$$

Answer:

$3e^{4x+2}+C$

Working:

Let $u=4x+2$.

Then $\dfrac{du}{dx}=4$, so $dx=\dfrac{1}{4}\,du$.

Substitute: $\int 12e^{4x+2}\,dx=\int 12e^u\cdot\dfrac{1}{4}\,du$.

Simplify: $\int 3e^u\,du$.

Integrate: $3e^u+C$.

Replace u with $4x+2$: $3e^{4x+2}+C$.

Teacher answers & working

Integration by Substitution and Parts · Integration by Substitution

ALG-99DB-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S4

Integrate each expression using a linear substitution.

$\int 20(5x+3)^4\,dx$

Answer:

$\frac{4}{5}(5x+3)^5 + C$

Working:

Let $u = 5x + 3$.

Differentiate: $\frac{du}{dx} = 5$, so $du = 5\,dx$ and $dx = \frac{1}{5}\,du$.

Substitute into the integral: $\int 20(5x+3)^4\,dx = \int 20u^4 \cdot \frac{1}{5}\,du$.

Simplify the constant: $\int 4u^4\,du$.

New power: $4 + 1 = 5$.

Integrate: $4 \cdot \frac{u^5}{5} + C$.

Simplify the coefficient: $4/5 = \frac{4}{5}$.

$\frac{4}{5}u^5 + C$.

Replace u with $5x + 3$: $\frac{4}{5}(5x+3)^5 + C$.

Teacher answers & working

Integration by Substitution and Parts · Integration by Substitution

ALG-99DB-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EA1

Integrate each expression using a quadratic substitution.

$$\int 3(4x-5)(2x^2-5x-1)^3\,dx$$

Answer:

$$\frac{3}{4}(2x^2-5x-1)^4+C$$

Working:

Let $u=2x^2-5x-1$.

Differentiate: $\frac{du}{dx}=4x-5$, so $du=(4x-5)\,dx$.

The factor $(4x-5)\,dx$ appears in the integral, so replace it with du.

The integral becomes $\int 3u^3\,du$.

New power: $3+1=4$.

Integrate: $3\cdot\frac{u^4}{4}+C$.

Simplify the coefficient: $3/4=\frac{3}{4}$.

$\frac{3}{4}u^4+C$.

Replace u with the original expression: $\frac{3}{4}(2x^2-5x-1)^4+C$.

EA2

Integrate each exponential expression using substitution.

$$\int 9e^{3x-5}\,dx$$

Answer:

$3e^{3x-5}+C$

Working:

Let $u=3x-5$.

Then $\frac{du}{dx}=3$, so $dx=\frac{1}{3}\,du$.

Substitute: $\int 9e^{3x-5}\,dx=\int 9e^u\cdot\frac{1}{3}\,du$.

Simplify: $\int 3e^u\,du$.

Integrate: $3e^u+C$.

Replace u with $3x-5$: $3e^{3x-5}+C$.

Teacher answers & working

Integration by Substitution and Parts · Integration by Substitution

ALG-99DB-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EB1

Integrate each expression using a linear substitution.

$$\int 25(5x+7)^5\,dx$$

Answer:

$$\frac{5}{6}(5x+7)^6+C$$

Working:

Let $u = 5x + 7$.

Differentiate: $\frac{du}{dx} = 5$, so $du = 5\,dx$ and $dx = \frac{1}{5}\,du$.

Substitute into the integral: $\int 25(5x+7)^5\,dx = \int 25u^5 \cdot \frac{1}{5}\,du$.

Simplify the constant: $\int 5u^5\,du$.

New power: $5 + 1 = 6$.

Integrate: $5 \cdot \frac{u^6}{6} + C$.

Simplify the coefficient: $5/6 = \frac{5}{6}$.

$\frac{5}{6}u^6 + C$.

Replace u with $5x + 7$: $\frac{5}{6}(5x+7)^6 + C$.